Mathematics - Undergraduate Level Assessment 24 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. Find all zeros in the indicated finite field of the given polynomial with coefficients in that field. x^3 + 2 x + 2 in Z_7
   (a) 1
   (b) 2
   (c) 2,3
   (d) 6
2. Using Fermat's theorem, find the remainder of $3^4$7 when it is divided by 23.
   (a) 1
   (b) 2
   (c) 3
   (d) 4
3. Statement 1 | A homomorphism may have an empty kernel. Statement 2 | It is not possible to have a nontrivial homomorphism of some finite group into some infinite group.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
4. Statement 1 | Q is an extension field of Z_2. Statement 2 | Every non-constant polynomial over a field has a zero in some extension field.
   (a) True, True
   (b) False, False
   (c) True, False
   (d) False, True
5. If f(z) is an analytic function that maps the entire finite complex plane into the real axis, then the imaginary axis must be mapped onto
   (a) the entire real axis
   (b) a point
   (c) a ray
   (d) an open finite interval

True / False (10 marks)

Answer True or False

6. True or False: The order of a finite group containing a subgroup of order five and no element other than the identity as its own inverse could be 35.
7. True or False: The polynomial 8 x^3 + 6 x^2 - 9 x + 24 satisfies the Eisenstein criterion for irreducibility with p=3.
8. True or False: If R is a ring and f(x) and g(x) are in R[x], then deg(f(x) + g(x)) = deg(f(x)) + deg(g(x)).
9. True or False: The degree of the field extension Q($\sqrt{}$ 2, $\sqrt{}$ 3) over Q is 2.
10. True or False: Every ring necessarily has a multiplicative identity, which is not true for all rings.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the validity of the statements regarding subgroups and group order: Explain how the formula |HK| = |H||K|/|H K| applies, and analyze the structure of groups of order 2 p.
12. Given that the characteristic polynomial of a 3 × 3 matrix A is det(A - I) = -$^3 + 3^2 + -3, analyzetheimplicationsforthetrace, determinant, andeigenvaluesofA.$

End of Paper